IMPROVED KISSING NUMBERS IN TWENTY-EIGHT THROUGH THIRTY-ONE DIMENSIONS

Abstract. We prove that the kissing numbers in 28, 29, 30 and 31 dimensions are at least 205172, 210495, 222073 and 241079. The best previously published lower bounds, due to Takhanov and Yun, were 204522, 209497, 220442 and 238354. As in the constructions of Kallal, Kan and Wang and of Ma et al., we lift part of the Leech lattice kissing configuration into the extra coordinates, replacing some minimal vectors by two or three points each. The replaced vectors come in sets with pairwise cosines at most 1/4, and one more vector in every set is worth 16, 26, 48 or 84 points. We construct such sets of 528 vectors, where Ma et al. used 496, from the cosets of a five-dimensional subcode of the binary Golay code that contains no octad, together with a perfect matching of an octad. This change alone beats all previous bounds. In dimension 28 we then replace the points in the extra coordinates by cross-polytopes of Leech lattice vectors, with a proof by hand apart from finite checks on printed data. In dimensions 29 to 31 we add points found by numerical optimization and check every pair in exact arithmetic.

1. Introduction

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [3, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum and call such a set a *kissing configuration*. In dimension 24 the 196560 minimal vectors of the Leech lattice attain τ_{24} [7, 11].

For $25 \le n \le 31$ the best constructions start from this configuration. Cohn, Jiao, Kumar and Torquato [1] and Kallal, Kan and Wang [4] remove some minimal vectors and put each of them back as two or three points with a short component in the extra $n-24$ coordinates, then add points that lie in the extra coordinates alone. Ma et al. [8] state this construction in general and found, by a game-theoretic search, configurations of 204520, 209496, 220440 and 238350 points in dimensions 28 to 31. Takhanov and Yun [12] rotate parts of these configurations and gain a few more points, reaching 204522, 209497, 220442 and 238354. Kravatskiy's public repository [5] lists the larger values 204896, 209968, 221012 and 238662, which we have not checked.

Theorem 1.1. *$\tau_{28} \ge 205172$, $\tau_{29} \ge 210495$, $\tau_{30} \ge 222073$ and $\tau_{31} \ge 241079$.*

The idea. We call the construction of [1, 4, 8] a *Leech lift* (Section 2). Its input is a family of pairwise disjoint *head sets*, sets of Leech minimal vectors with pairwise cosines at most 1/4, and for each head set a *tail set* of two or three unit vectors in the extra coordinates. Each vector of a head set is replaced by one point per tail, so a head set S_i with tail set T_i contributes $(|T_i|-1)|S_i|$ points beyond the 196560. The tail sets of [8] are triangles and antipodal pairs of roots of D_4, D_5, E_6 and E_7. A tail set has at most three tails, so a set $\mathcal{T}$ of tails gives at most $\lfloor 2|\mathcal{T}|/3 \rfloor$ for $\sum_i (|T_i|-1)$, and these root systems reach this with 16, 26, 48 and 84 (Lemma 2.2); in dimension 4 nothing better is possible, since $\tau_4 = 24$ [10]. So one more vector

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TABLE 1. Lower and upper bounds in dimensions 28 to 31. The published lower bounds are those of Takhanov and Yun. The values of Kravatskiy's repository are not published in a paper, and we have not checked them. The upper bounds are those of Cohn's table [2]: floors of numerical semidefinite programming values [6], not interval-certified.

dimension	Takhanov and Yun [12]	Kravatskiy [5]	this paper	upper bound
28	204522	204896	**205172**	752292
29	209497	209968	**210495**	1075991
30	220442	221012	**222073**	1537707
31	238354	238662	**241079**	2213487

in every head set is worth 16, 26, 48 or 84 points. Ma et al. use head sets of 496 vectors. Ours have 528.

Here is the construction. Index the coordinates of $\mathbb{R}^{24}$ by the 24 points of the Golay code G, and fix an octad O and a five-dimensional subcode $K \leq G$ that contains no octad. At each of the 16 positions outside O take the 32 minimal vectors of shape $(\mp 3, \pm 1^{23})$ whose entry ∓ 3 is at that position and whose sign pattern runs over a coset of K. Two of them at the same position have cosine at most $1/4$ because K has no octad, and a good choice of the 16 cosets keeps different positions apart. All 512 vectors are ± 1 on O, so the 16 vectors $(\pm 4, \pm 4, 0^{22})$ supported on the pairs of a perfect matching of O can be added (Section 3). Translating the cosets gives disjoint copies, and matchings with no common pair give disjoint copies on the same octad. With these head sets and no points at all in the extra coordinates, the Leech lift already has 205008, 210288, 221904 and 240912 points, more than any previous bound (Corollary 3.4).

The rest of the gain comes from the extra coordinates, where Ma et al. place 24, 40, 72 and 126 *poles*, points of the form $(0, w)$. A pole is orthogonal to the whole body, so the head directions near it are unused. In dimension 28 we replace a pole by eight points: for a set R of four coordinates, the eight vectors 4σ, with σ a sign vector of one parity on R, lie in the Leech lattice and form a cross-polytope, and we tilt this cross-polytope by 45° toward the pole, taking the points proportional to $(4\sigma, 8w)$. They have pairwise cosines 0 or $\frac{1}{2}$, and they fit the rest of the configuration as long as every head set lifted near the direction w meets each 4σ in at most 16, which holds when the head set lies on an octad through R and its matching separates R (Lemma 4.1). Twenty such cross-polytopes and four poles give 164 points where Ma et al. have 24, and the proof is by hand (Section 4).

In dimensions 29 to 31 we remove all poles and add 207, 169 and 167 points: 144 layer points and 63 computed points in dimension 29, 72 hole points and 97 computed points in dimension 30, and 167 computed points in dimension 31. The computed points were found by numerical optimization against the rest of the configuration and rounded to integer vectors, and every pair that contains one of them is checked in exact arithmetic (Section 5). Table 2 shows how the gains over [8] split.

The proofs are by hand except for finite facts about explicit sets, checked by computer in exact arithmetic; we name each where it is used and list them in Section 6. The accompanying data list every point of each configuration, and an accompanying program checks all pairs of each list exactly. Section 2 states the Leech lift, Section 3 builds the head sets, and Sections 4 and 5 give the configurations.

TABLE 2. The gain over the configurations of Ma et al. [8], split into the gain from the 528-point head sets alone (with no points in the extra coordinates) and the points added in the extra coordinates.

dimension	Ma et al. [8]	head sets alone	extra coordinates	total	gain
28	204520	205008	164	205172	652
29	209496	210288	207	210495	999
30	220440	221904	169	222073	1633
31	238350	240912	167	241079	2729

2. The Leech lift

Let $\Lambda \subset \mathbb{R}^{24}$ be the Leech lattice, scaled so that its minimal vectors have squared norm 32 and integer coordinates [3, Ch. 4, §11], and let $\Lambda_{\min}$ be the set of its 196560 minimal vectors. At this scale all inner products in Λ are multiples of 8, and distinct vectors of Λ are at squared distance at least 32, so if $x, y \in \Lambda$ are distinct, then

$$\langle x, y\rangle \le \tfrac{1}{2}\big(|x|^2 + |y|^2 - 32\big). \tag{2.1}$$

In particular, inner products of distinct vectors of $\Lambda_{\min}$ lie in $\{0, \pm 8, \pm 16, -32\}$.

For nonzero x, y the angle between them is at least 60° if and only if

$$\langle x, y\rangle \le 0 \quad \text{or} \quad 4\langle x, y\rangle^2 \le |x|^2|y|^2. \tag{2.2}$$

We call such a pair *compatible*. If every pair from a finite set of nonzero vectors is compatible, then the vectors normalized to unit length form a kissing configuration, and distinct vectors give distinct points because (2.2) excludes cosine 1.

Write $n = 24 + d$. We call the first 24 coordinates of $\mathbb{R}^n$ the *head* and the last d the *tail*. A *head set* is a subset $S \subseteq \Lambda_{\min}$ with $\langle u, u'\rangle \le 8$ for all distinct $u, u' \in S$, that is, with pairwise cosines at most $1/4$. Put $\alpha = \sqrt{2/3}$ and $\beta = \sqrt{1/3}$, and for $x \in \Lambda_{\min}$ let $\hat{x} = x/\sqrt{32}$. The data of a Leech lift are

- pairwise disjoint head sets $S_1, \dots, S_k$;
- *tail sets* $T_1, \dots, T_k$ of unit vectors in $\mathbb{R}^d$, with $\langle t, t'\rangle \le -\frac{1}{2}$ for distinct t, t' in one tail set and $\langle t, t'\rangle \le \frac{1}{2}$ for t, t' in different tail sets;
- a set P of *poles*, unit vectors in $\mathbb{R}^d$ with $\langle p, p'\rangle \le \frac{1}{2}$ for distinct poles and $\langle t, p\rangle \le \frac{\sqrt{3}}{2}$ for every tail t and pole p.

The configuration is $X = B \cup L \cup Z$ with

$$B = \{(\hat{b}, 0) : b \in \Lambda_{\min} \setminus \textstyle\bigcup_i S_i\} \qquad \text{(body)},$$
$$L = \{(\alpha\hat{h}, \beta t) : h \in S_i,\ t \in T_i,\ 1 \le i \le k\} \qquad \text{(lifts)},$$
$$Z = \{(0, p) : p \in P\} \qquad \text{(poles)}.$$

This is the construction of [8, Proposition 5] in our normalization.

Proposition 2.1. *X is a kissing configuration in $\mathbb{R}^{24+d}$ of size*

$$196560 + \sum_{i=1}^{k} (|T_i| - 1)\,|S_i| + |P|. \tag{2.3}$$

Proof. All points are unit vectors, and we bound the inner product of two distinct points. Two body points: $\langle b, b'\rangle/32 \le \frac{1}{2}$. A body point and a lift: $\alpha\langle b, h\rangle/32 \le \alpha/2$, since $b \ne h$. Two lifts

$(\alpha\hat{h}, \beta t)$, $(\alpha\hat{h}', \beta t')$ with $h \in S_i$, $h' \in S_j$ have inner product $\frac{2}{3} \cdot \frac{\langle h,h' \rangle}{32} + \frac{1}{3}\langle t, t' \rangle$. If $h = h'$, then $t \neq t'$ lie in one tail set and the value is at most $\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$. If $h \neq h'$ and $i = j$, then $\langle h, h' \rangle \leq 8$ and the value is at most $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$. If $i \neq j$, then $\langle h, h' \rangle \leq 16$ and $\langle t, t' \rangle \leq \frac{1}{2}$, so the value is at most $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$. A lift and a pole: $\beta\langle t, p \rangle \leq \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2}$. Two poles: $\langle p, p' \rangle \leq \frac{1}{2}$. A body point and a pole: 0. The count is $(196560 - \sum |S_i|) + \sum |S_i||T_i| + |P|$. □

The proposition holds for every assignment of head sets to tail sets. The points added in Sections 4 and 5 do depend on the assignment, and we choose it there.

Lemma 2.2. *Every tail set has at most three tails, and if all tails together form a set $\mathcal{T}$, then $\sum_i (|T_i| - 1) \leq \lfloor 2|\mathcal{T}|/3 \rfloor \leq \lfloor 2\tau_d/3 \rfloor$.*

Proof. Unit vectors $t_1, \ldots, t_m$ with pairwise inner products at most $-\frac{1}{2}$ satisfy $0 \leq |\sum t_j|^2 \leq m - m(m-1)/2$, so $m \leq 3$. All tails together form a kissing configuration in $\mathbb{R}^d$, so $|\mathcal{T}| \leq \tau_d$, and $|T_i| - 1 \leq \frac{2}{3}|T_i|$. □

The tails. We use the tails of [8]. Scaled to squared norm 2 they are the roots of a root system of type D_4, D_5, E_6 and E_7 in dimensions 28 to 31, with 24, 40, 72 and 126 roots [3, Ch. 4]. Distinct roots of a simply laced root system have cosines in $\{0, \pm\frac{1}{2}, -1\}$, so any partition of the roots into *triangles* $\{r, s, -r - s\}$ with $\langle r, s \rangle = -\frac{1}{2}$ and antipodal pairs satisfies the tail set conditions. We use 8 triangles in dimension 28, 12 triangles and 2 pairs in dimension 29, 24 triangles in dimension 30 and 42 triangles in dimension 31; the partitions are listed with the accompanying data. Then $\sum_i (|T_i| - 1) = 16$, 26, 48 and 84, which is $\lfloor 2|\mathcal{T}|/3 \rfloor$, so no other partition of these tails does better. A larger set of tails would need a kissing configuration in dimension 5, 6 or 7 larger than the roots of D_5, E_6 or E_7, which are the largest known; in dimension 4 the 24 roots of D_4 are optimal [10], and there (2.3) can only grow through the head sets and the points in the extra coordinates.

3. Head sets of 528 vectors

The Golay code in the shell. Index the coordinates of $\mathbb{R}^{24}$ by $\{0, 1, \ldots, 23\}$ and identify binary words of length 24 with subsets of this set. The vectors of $\Lambda_{\min}$ have three shapes [3, Ch. 4, §11]: 1104 of shape $(\pm 4, \pm 4, 0^{22})$, 97152 of shape $(\pm 2^8, 0^{16})$ and 98304 of shape $(\mp 3, \pm 1^{23})$. For a position i and a word a, let $v(i, a)$ be the vector with entry -3 at i and 1 elsewhere, with the signs of the coordinates in a changed. Let G be the extended binary Golay code spanned by the twelve octads

$$\{0,1,2,3,4,11,19,21\},\ \{0,1,2,3,5,15,20,23\},\ \{0,1,2,3,6,9,10,12\},\ \{0,1,2,3,7,13,18,22\},$$
$$\{0,1,2,3,8,14,16,17\},\ \{0,1,2,4,5,9,13,17\},\ \{0,1,2,4,6,16,22,23\},\ \{0,1,2,4,7,8,12,20\},$$
$$\{0,1,2,5,6,8,11,18\},\ \{0,1,2,5,7,10,16,21\},\ \{0,1,3,4,5,8,10,22\},\ \{0,2,3,4,5,6,7,14\};$$

all words printed in this paper are words of this G. A vector with odd entries lies in Λ if and only if its coordinate sum is 4 modulo 8 and its coordinates congruent to 3 modulo 4 form a word of G. For $v(i, a)$ these coordinates are exactly a, and the coordinate sum is $20 - 2|a|$ or $28 - 2|a|$, which is 4 modulo 8 for every $a \in G$ because all weights in G are divisible by 4. So the vectors of shape $(\mp 3, \pm 1^{23})$ are exactly the $24 \cdot 4096$ vectors $v(i, a)$ with $a \in G$, and the supports of the vectors of shape $(\pm 2^8, 0^{16})$ are the 759 octads of G.

Lemma 3.1 (conflict rule)**.** *Let $a, b \in G$ and $c = a + b$. Then $\langle v(i, a), v(j, b) \rangle = 16$ in exactly these cases: $i = j$ and c is an octad not containing i; $i \neq j$ and $c = 0$; $i \neq j$ and c is an octad containing i and j. In all other cases with $v(i, a) \neq v(j, b)$ the inner product is at most 8.*

Proof. Let $w = \mathrm{wt}(c)$ and $\epsilon_k = -1$ if $k \in c$, $\epsilon_k = 1$ otherwise, so that $\sum_k \epsilon_k = 24 - 2w$. The two vectors agree in sign exactly off c. If $i = j$, the inner product is $\sum_{k \neq i} \epsilon_k + 9\epsilon_i = 24 - 2w + 8\epsilon_i$. If $i \neq j$, it is $\sum_{k \neq i,j} \epsilon_k - 3\epsilon_i - 3\epsilon_j = 24 - 2w - 4\epsilon_i - 4\epsilon_j$. The weights in G are $0, 8, 12, 16, 24$, and the values follow. □

The construction. A subcode $K \leq G$ that contains no octad has all nonzero weights in $\{12, 16, 24\}$, so its dimension is at most 5: a binary $[24, 6, 12]$ code would violate the Griesmer bound $24 \geq 12 + 6 + 3 + 2 + 1 + 1$ [9, Ch. 17, §5]. By Lemma 3.1 with $i = j$, the 32 vectors $v(i, a)$ with a in a coset of a five-dimensional octad-free K have pairwise inner products at most 8. By the same lemma, vectors at two different positions i, j conflict exactly when the sum of their words is 0 or an octad through i and j, and the following definition asks for cosets at the 16 positions outside an octad O with no such sum.

Definition 3.2. Let $K \leq G$ be a five-dimensional subcode with no octad, and let O be an octad. A *coset assignment* for (K, O) is a map $i \mapsto \gamma_i \in G/K$ on the 16 positions $i \notin O$ such that for all $i \neq j$ outside O the coset $\gamma_i + \gamma_j$ contains neither 0 nor an octad through i and j. For a coset assignment γ and a perfect matching M of O (four disjoint pairs covering O) let

$$H(\gamma, M) = \{v(i, a) : i \notin O,\ a \in \gamma_i\} \ \cup \ F(M),$$

where $F(M)$ is the set of the 16 vectors $(\pm 4, \pm 4, 0^{22})$ supported on the pairs of M.

Theorem 3.3. *$H(\gamma, M)$ is a head set of $16 \cdot 32 + 16 = 528$ vectors. If $c \in G/K$ is nonzero and M' is a perfect matching of O with no pair in common with M, then $H(\gamma + c, M')$ is a head set disjoint from $H(\gamma, M)$; here $\gamma + c$ is the assignment $i \mapsto \gamma_i + c$.*

Proof. Two vectors $v(i, a)$, $v(j, b)$ of $H(\gamma, M)$ have inner product at most 8: for $i = j$ by Lemma 3.1, since $a + b \in K$ is not an octad, and for $i \neq j$ by Lemma 3.1 and the condition on $\gamma_i + \gamma_j$. Every $v(i, a)$ has entries ± 1 on O, so it meets a vector $4(\pm e_p \pm e_q)$ with $p, q \in O$ in $4(\pm 1 \pm 1) \in \{0, \pm 8\}$. Two vectors of $F(M)$ on different pairs have disjoint supports, and two on the same pair have inner product 0 or -32. The condition on $\gamma_i + \gamma_j$ is unchanged by adding c to every γ_i, so $\gamma + c$ is a coset assignment. The two head sets are disjoint because they use different cosets at every position and $F(M) \cap F(M')$ consists of the vectors on common pairs. □

For comparison, the head sets of [8] have 496 vectors: in their coordinates, three pieces of 128, 128 and 192 vectors lying in the three 16-dimensional coordinate subspaces of a trio of octads, and 48 vectors of shape $(\pm 4, \pm 4, 0^{22})$ on the pairs of a perfect matching of all 24 coordinates. Ours take 512 vectors of the single shape $(\mp 3, \pm 1^{23})$, with the entry ∓ 3 outside one octad, and 16 vectors of shape $(\pm 4, \pm 4, 0^{22})$ inside it. We do not know the largest head set.

The subcodes and the assignments. We use five octad-free subcodes $K_1, \ldots, K_5$ of dimension 5, each with weight distribution $0^1\, 12^{28}\, 16^3$. The complements of the three words of weight 16 in K_f are three pairwise disjoint octads, a *trio*; we write them O_{f1}, O_{f2}, O_{f3}. Table 3 lists a basis of each subcode and its trio. For each pair (K_f, O) that we use, an exhaustive search over G/K_f found a coset assignment, and we use its translates: by Theorem 3.3, translates by distinct cosets, on matchings with no common pair, are pairwise disjoint. The assignments for dimension 28 are printed in Section 4; those for dimensions 29 to 31 are given with the accompanying data. That each printed or listed assignment satisfies Definition 3.2 is a finite check by computer.

TABLE 3. The five subcodes. Each row gives a basis: two words of weight 16 and three of weight 12. The trio consists of the complements of the weight-16 words of K_f.

K_1	$\{0,1,3,5,6,7,8,10,12,13,14,15,16,18,19,20\}$, $\{2,4,5,6,8,9,11,12,13,15,16,17,18,21,22,23\}$, $\{0,1,2,3,5,8,11,13,15,17,19,22\}$, $\{0,1,2,4,6,8,9,10,13,18,20,22\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$
trio	$O_{11} = \{0,1,3,7,10,14,19,20\}$, $O_{12} = \{2,4,9,11,17,21,22,23\}$, $O_{13} = \{5,6,8,12,13,15,16,18\}$
K_2	$\{0,1,3,4,6,7,10,12,14,16,17,18,19,20,21,22\}$, $\{2,3,4,5,7,8,9,11,12,13,14,15,16,17,20,23\}$, $\{0,1,2,3,4,5,7,10,12,13,15,19\}$, $\{0,1,2,3,9,11,12,13,14,17,21,22\}$, $\{0,2,3,4,5,6,9,10,17,20,22,23\}$
trio	$O_{21} = \{0,1,6,10,18,19,21,22\}$, $O_{22} = \{2,5,8,9,11,13,15,23\}$, $O_{23} = \{3,4,7,12,14,16,17,20\}$
K_3	$\{0,2,4,5,7,8,9,10,11,13,14,18,19,20,21,22\}$, $\{1,2,3,4,6,7,10,11,12,14,15,16,17,20,21,23\}$, $\{0,1,2,3,4,5,8,9,11,12,14,15\}$, $\{0,1,2,3,4,8,13,17,20,21,22,23\}$, $\{0,1,2,5,6,7,14,15,18,21,22,23\}$
trio	$O_{31} = \{0,5,8,9,13,18,19,22\}$, $O_{32} = \{1,3,6,12,15,16,17,23\}$, $O_{33} = \{2,4,7,10,11,14,20,21\}$
K_4	$\{0,1,2,3,4,5,6,9,10,12,13,16,17,18,19,20\}$, $\{0,1,4,7,8,10,11,12,14,15,16,17,19,21,22,23\}$, $\{5,6,7,9,10,11,12,14,15,16,18,19\}$, $\{1,2,3,6,7,8,9,10,15,16,17,21\}$, $\{1,2,4,6,7,8,11,12,13,16,18,22\}$
trio	$O_{41} = \{0,1,4,10,12,16,17,19\}$, $O_{42} = \{2,3,5,6,9,13,18,20\}$, $O_{43} = \{7,8,11,14,15,21,22,23\}$
K_5	$\{0,1,4,5,6,9,10,11,12,15,16,17,18,19,21,22\}$, $\{0,1,2,3,5,7,8,9,10,11,13,14,15,19,20,23\}$, $\{0,1,4,7,8,9,11,12,13,14,16,17\}$, $\{2,4,6,7,9,10,11,13,16,18,19,20\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$
trio	$O_{51} = \{0,1,5,9,10,11,15,19\}$, $O_{52} = \{4,6,12,16,17,18,21,22\}$, $O_{53} = \{2,3,7,8,13,14,20,23\}$

Disjoint head sets. An octad has $\binom{8}{2} = 28$ pairs and a matching uses 4, so at most 7 of our head sets lie on one octad, and one trio carries at most 21. Two distinct octads share 0, 2 or 4 points, and a shared pair can be used by only one matching; the six octads of the trios of K_1 and K_2 contain 140 distinct pairs, which allows at most 35 matchings, and the nine octads of K_1, K_2, K_3 contain 177. This is why dimension 30 uses two subcodes and dimension 31 three. Two 512-point parts on different octads, or from different subcodes K_f, K_g, with assignments γ, γ' and translation labels c, c', meet exactly when $\gamma_i + \gamma'_i + c + c'$ lies in $K_f + K_g$ for some position i outside both octads. For K_1, K_2, K_3 the three sums $K_f + K_g$ all equal $K_1 + K_2 + K_3$, of dimension 8, so the labels can be chosen modulo this subcode. The labels are listed with the accompanying data, and we checked by computer that the head sets in each dimension are pairwise disjoint. Table 4 says how many lie on each octad.

TABLE 4. Number of head sets on each octad.

dimension	O_{11}	O_{12}	O_{13}	O_{21}	O_{22}	O_{23}	O_{31}	O_{32}	O_{33}	O_{41}	O_{51}	total
28	4	–	–	4	–	–	–	–	–	–	–	8
29	4	–	–	4	–	–	–	–	–	4	2	14
30	–	6	5	6	–	7	–	–	–	–	–	24
31	6	7	6	4	3	3	4	5	4	–	–	42

Corollary 3.4. *With* 8, 14, 24 *and* 42 *pairwise disjoint head sets of 528 vectors on the tail sets of Section 2 and no poles, the Leech lift is a kissing configuration of* $196560+528\cdot16 = 205008$, $196560 + 528 \cdot 26 = 210288$, 221904 *and* 240912 *points in dimensions 28 to 31.*

Proof. Proposition 2.1 with $P = \emptyset$. □

4. Dimension 28: cross-polytopes in place of poles

Head sets and a tetrad. A *tetrad* is a set of four coordinates. Every 5-set lies in exactly one octad, since the octads form a Steiner system $Q(5, 8, 24)$ [3, Ch. 10, §2]. So a tetrad R lies in exactly five octads, and two of them meet exactly in R: each of the 20 points outside R lies in exactly one octad through R. For $\sigma \in \{\pm1\}^R$ the vector 4σ is the sum of two vectors of shape $(\pm4, \pm4, 0^{22})$ with disjoint supports, so it lies in Λ, with squared norm 64. We say a matching M of an octad $O \supseteq R$ is *R-bipartite* if every pair of M has exactly one point in R. The R-bipartite pairs form a complete bipartite graph $K_{4,4}$, whose 16 edges split into four R-bipartite matchings with no common pair.

Lemma 4.1. *Let $R \subseteq O$ be a tetrad in an octad, let A be a set of vectors of shape $(\mp3, \pm1^{23})$ whose entries ∓3 lie outside O, and let M be an R-bipartite matching of O. Then $|\langle h, 4\sigma\rangle| \le 16$ for every $h \in A \cup F(M)$ and every $\sigma \in \{\pm1\}^R$. If instead M has a pair inside R, then $\langle h, 4\sigma\rangle = 32$ for some $h \in F(M)$ and some σ.*

Proof. A vector of A has entries ±1 on R, so it meets 4σ in $4\sum_{k\in R}\pm1$, of absolute value at most 16. A vector $4(\pm e_p \pm e_q)$ with exactly one of p, q in R meets 4σ in ±16. If $p, q \in R$, choosing the signs to agree gives 32. □

When the entry ∓3 of a head lies inside R, the inner product with 4σ can reach 24. The following parity rule says when it does not.

Lemma 4.2. *Let $h = v(i, a)$ with $i \in R$ and let $\sigma \in \{\pm1\}^R$ have m minus signs. Then $\langle h, 4\sigma\rangle \le 16$ if $|a \cap R| \equiv m \pmod 2$, and $\langle h, 4\sigma\rangle \le 24$ in any case.*

Proof. On R the vector h has entry $-3s_i$ at i and s_k at the other three points, where $s_k = -1$ for $k \in a$ and $s_k = 1$ otherwise. Put $x_k = s_k\sigma_k$, so that $\langle h, 4\sigma\rangle = 4(-3x_i + \sum_{k\ne i} x_k)$. The sum in parentheses is at most 6, and equals 6 only when $x_i = -1$ and the other three x_k are 1, where $\prod_{k\in R} x_k = -1$. If $|a \cap R| \equiv m$, then $\prod x_k = (-1)^{|a\cap R|}(-1)^m = 1$, so the value 6 is excluded, and the next possible value is 4, which gives $\langle h, 4\sigma\rangle \le 16$. □

Layer points. Let $v \in \Lambda$ with $|v|^2 = 64$ and let $w \in \mathbb{R}^d$ be a unit vector. The *layer point* with data (v, w) is the unit vector

$$Y(v, w) = \Big(\frac{v}{8\sqrt{2}},\ \frac{w}{\sqrt{2}}\Big),$$

proportional to $(v, 8w)$. All our layer points have $v = 4\sigma$ with σ a sign vector on a tetrad R. Two sign vectors on R that differ in j places have $\langle 4\sigma, 4\sigma'\rangle = 16(4 - 2j)$: distinct sign vectors of the same parity give 0 or -64, and different parities give ±32. Sign vectors on tetrads R, R' have $|\langle 4\sigma, 4\sigma'\rangle| \le 16|R \cap R'|$. For a fixed w and R, the eight vectors 4σ with σ of one parity are the vertices of a cross-polytope in $\mathbb{R}^R$, and the eight layer points $Y(4\sigma, w)$ have pairwise inner products $\frac12 + \langle 4\sigma, 4\sigma'\rangle/128$, which is $\frac12$ or 0: they are that cross-polytope tilted by 45° toward the direction $(0, w)$ of a pole.

Lemma 4.3. *Let $Y = Y(v, w)$ be a layer point.*

(1) $\langle Y, (\hat{b}, 0)\rangle = \langle v, b\rangle/64 \le \frac12$ *for every $b \in \Lambda_{\min}$.*

(2) $\langle Y, (\alpha\hat{h}, \beta t)\rangle = \dfrac{\langle h, v\rangle}{32\sqrt{6}} + \dfrac{\langle t, w\rangle}{\sqrt{6}}$. *This is at most* $\frac{1}{2}$ *if* $\langle t, w\rangle \le 0$, *and also if* $\langle h, v\rangle \le 16$ *and* $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$.

(3) $\langle Y, (0, p)\rangle = \langle w, p\rangle/\sqrt{2}$.

Proof. The formulas are direct. (1) is (2.1). In (2), (2.1) gives $\langle h, v\rangle \le 32$, so $\langle t, w\rangle \le 0$ gives at most $\frac{1}{\sqrt{6}}$; and $\langle h, v\rangle \le 16$, $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$ give at most $\frac{16}{32\sqrt{6}} + \frac{1}{\sqrt{12}} < \frac{1}{2}$. □

Lemma 4.4. *Let* $Y = Y(4\sigma, w)$ *and* $Y' = Y(4\sigma', w')$ *be distinct layer points on tetrads* R, R'. *Then* $\langle Y, Y'\rangle = \langle 4\sigma, 4\sigma'\rangle/128 + \langle w, w'\rangle/2$, *and this is at most* $\frac{1}{2}$ *in each of the following cases: (P1)* $w = w'$, $R = R'$ *and* σ, σ' *have the same parity; (P2)* $\langle w, w'\rangle \le 0$; *(P3)* $\langle w, w'\rangle \le \frac{1}{2}$ *and* $|\langle 4\sigma, 4\sigma'\rangle| \le 32$, *which holds when* $R = R'$ *and the parities differ, and when* $|R \cap R'| \le 2$.

Proof. In (P1) $\sigma \ne \sigma'$ have the same parity, so $\langle 4\sigma, 4\sigma'\rangle \le 0$. In (P2) the first term is at most $\frac{64}{128}$. In (P3) the value is at most $\frac{32}{128} + \frac{1}{4}$. □

The tails. The 24 tails are the roots of D_4, scaled to unit length. Choose an orthonormal basis $e_1, \dots, e_4$ of $\mathbb{R}^4$ in which they are $(\pm e_i \pm e_j)/\sqrt{2}$. The 24 unit vectors

$$\Pi_0 = \{\pm e_i\}, \qquad \Pi_+ = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an even number of minus signs}\},$$
$$\Pi_- = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an odd number of minus signs}\}$$

point at the deep holes of the D_4 root lattice. Each of Π_0, Π_+, Π_- is the vertex set of a cross-polytope: two of its vectors have inner product 0 or -1. Vectors of different classes have inner product $\pm\frac{1}{2}$. A tail has inner product 0 or $\pm\frac{1}{\sqrt{2}}$ with each of the 24 vectors.

Lemma 4.5. *Let* $c \in \Pi_0$. *Of the eight triangles of tails, exactly two are perpendicular to* c, *and each of the other six has exactly one tail* t *with* $\langle t, c\rangle = \frac{1}{\sqrt{2}}$.

Proof. For a triangle $\{r, s, -r - s\}$ the values $\langle r, c\rangle$, $\langle s, c\rangle$, $-\langle r, c\rangle - \langle s, c\rangle$ lie in $\{0, \pm\frac{1}{\sqrt{2}}\}$ and sum to 0. So either all are 0, or they are $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ in some order. Twelve of the 24 tails are not perpendicular to c, so six triangles are of the second kind and two of the first. □

The head sets. Let $Q = \{0, 1, 10, 19\}$. The octads $O_{11} = Q \cup \{3, 7, 14, 20\}$ and $O_{21} = Q \cup \{6, 18, 21, 22\}$ contain Q. We use eight head sets $H_j = H(\gamma^{(j)}, M_j)$, $j = 1, \dots, 8$: four on (K_1, O_{11}) and four on (K_2, O_{21}), where the four assignments on each octad are translates of one another and $M_1, \dots, M_4$ and $M_5, \dots, M_8$ are the four Q-bipartite matchings of O_{11} and of O_{21}:

$$M_1 = \{\{0, 14\}, \{19, 20\}, \{1, 3\}, \{7, 10\}\}, \qquad M_2 = \{\{0, 20\}, \{14, 19\}, \{1, 7\}, \{3, 10\}\},$$
$$M_3 = \{\{0, 3\}, \{7, 19\}, \{1, 14\}, \{10, 20\}\}, \qquad M_4 = \{\{0, 7\}, \{3, 19\}, \{1, 20\}, \{10, 14\}\},$$
$$M_5 = \{\{0, 21\}, \{1, 22\}, \{6, 10\}, \{18, 19\}\}, \qquad M_6 = \{\{0, 22\}, \{1, 21\}, \{10, 18\}, \{6, 19\}\},$$
$$M_7 = \{\{0, 6\}, \{1, 18\}, \{10, 21\}, \{19, 22\}\}, \qquad M_8 = \{\{0, 18\}, \{1, 6\}, \{10, 22\}, \{19, 21\}\}.$$

The assignments and the translation labels are as follows. Cosets of K_f in G are written by a representative word, the one of least weight and then lexicographically least. The assignment

γ on (K_1, O_{11}) is

$$\gamma_2 = K_1, \qquad \gamma_4 = \{0,1,14,15,16,19,21,23\} + K_1,$$
$$\gamma_5 = \{0,5,6,7,11,15,16,23\} + K_1, \qquad \gamma_6 = \{0,1,2,4,5,9,13,17\} + K_1,$$
$$\gamma_8 = \{0,3,4,7,8,13,17,19\} + K_1, \qquad \gamma_9 = \{0,2,8,13,15,17,18,20\} + K_1,$$
$$\gamma_{11} = \{0,3,4,6,8,11,12,16\} + K_1, \qquad \gamma_{12} = \{0,1,3,7,10,14,19,20\} + K_1,$$
$$\gamma_{13} = \{0,1,7,10,12,15,22,23\} + K_1, \qquad \gamma_{15} = \{0,2,4,6,12,13,18,19\} + K_1,$$
$$\gamma_{16} = \{0,1,3,5,7,11,12,17\} + K_1, \qquad \gamma_{17} = \{0,1,7,11,15,18,20,21\} + K_1,$$
$$\gamma_{18} = \{0,1,3,4,12,13,14,23\} + K_1, \qquad \gamma_{21} = \{0,2,6,9,14,17,18,22\} + K_1,$$
$$\gamma_{22} = \{0,1,5,11,13,21,22,23\} + K_1, \qquad \gamma_{23} = \{0,1,3,4,5,8,10,22\} + K_1,$$

and the head sets $H_1, \ldots, H_4$ are $H(\gamma + c_j, M_j)$ with the translation labels

$$c_1 = 0, \qquad c_2 = \{0,1,7,13,16,17,20,23\},$$
$$c_3 = \{0,1,2,3,5,15,20,23\}, \qquad c_4 = \{0,1,7,8,11,16,19,22\}.$$

The assignment γ' on (K_2, O_{21}) is

$$\gamma'_2 = \{0,3,4,7,10,12,18,21\} + K_2, \qquad \gamma'_3 = \{0,1,3,9,17,19,22,23\} + K_2,$$
$$\gamma'_4 = \{0,2,5,6,10,13,15,22\} + K_2, \qquad \gamma'_5 = \{0,1,3,6,7,8,21,23\} + K_2,$$
$$\gamma'_7 = \{0,5,8,10,11,13,16,17\} + K_2, \qquad \gamma'_8 = \{0,1,7,9,12,13,19,21\} + K_2,$$
$$\gamma'_9 = \{0,1,4,6,8,9,14,19\} + K_2, \qquad \gamma'_{11} = \{0,3,7,11,13,14,16,21\} + K_2,$$
$$\gamma'_{12} = \{0,2,10,12,15,19,20,21\} + K_2, \qquad \gamma'_{13} = \{0,1,4,5,11,14,16,20\} + K_2,$$
$$\gamma'_{14} = \{0,1,3,5,7,11,12,17\} + K_2, \qquad \gamma'_{15} = \{0,1,7,11,15,18,20,21\} + K_2,$$
$$\gamma'_{16} = \{0,8,9,11,14,15,16,18\} + K_2, \qquad \gamma'_{17} = \{0,2,3,10,14,21,22,23\} + K_2,$$
$$\gamma'_{20} = \{0,1,3,8,12,15,18,19\} + K_2, \qquad \gamma'_{23} = \{0,2,4,5,18,20,21,22\} + K_2,$$

and $H_5, \ldots, H_8$ are $H(\gamma' + c'_j, M_j)$ with

$$c'_5 = \{0,1,4,10,12,16,17,19\}, \qquad c'_6 = \{0,1,2,3,5,15,20,23\},$$
$$c'_7 = \{0,1,7,13,16,17,20,23\}, \qquad c'_8 = 0.$$

All representatives other than γ_2 are octads. The eight head sets are pairwise disjoint: on one octad by Theorem 3.3; across the two octads, a common vector $v(i,a)$ would need i outside both octads and a in a coset of K_1 and in a coset of K_2 at once, which the labels exclude (a finite check), and a common vector of $F(M_j)$ and $F(M_{j'})$ would be supported on a pair inside $O_{11} \cap O_{21} = Q$, while Q-bipartite pairs are not inside Q. By Lemma 4.1, every head h of the eight sets has $|\langle h, 4\sigma\rangle| \le 16$ for all $\sigma \in \{\pm 1\}^Q$.

Two more tetrads enter: $R_1 = \{0,1,6,18\} \subset O_{21}$ and $R_2 = \{0,3,7,19\} \subset O_{11}$, with $|Q \cap R_1| = |Q \cap R_2| = 2$ and $|R_1 \cap R_2| = 1$. For $j = 1, \ldots, 8$ let

$$m_1(j) = \max\{\langle h, 4\rho\rangle : h \in H_j,\ \rho \in \{\pm 1\}^{R_1} \text{ with an even number of minus signs}\},$$

and define $m_2(j)$ in the same way with R_2 and an odd number of minus signs.

Lemma 4.6. *$m_1(j) = 16$ for $j \neq 7, 8$ and $m_1(7) = m_1(8) = 32$; $m_2(j) = 16$ for $j \neq 3, 4$ and $m_2(3) = m_2(4) = 32$.*

Proof. (a) For the sets on the octad containing the tetrad this is Lemma 4.1: M_5, M_6 are R_1-bipartite and M_7, M_8 have a pair inside R_1; M_1, M_2 are R_2-bipartite and M_3, M_4 have a pair inside R_2.

(b) $H_1, \dots, H_4$ against R_1. Their vectors of $F(M_j)$ are supported on Q-bipartite pairs of O_{11}, and $R_1 \cap O_{11} = \{0, 1\} \subseteq Q$, so each pair has at most one point in R_1 and the inner product with 4ρ is at most 16. A head $v(i, a)$ with $i \notin R_1$ has entries ± 1 on R_1, so the inner product is at most 16. The positions of R_1 outside O_{11} are 6 and 18. The five basis words of K_1 in Table 3 meet $R_1 = \{0, 1, 6, 18\}$ in $\{0, 1, 6, 18\}$, $\{6, 18\}$, $\{0, 1\}$, $\{0, 1, 6, 18\}$ and $\{0, 6\}$, so every word of K_1 meets R_1 evenly; the representatives of γ_6 and γ_{18} and the labels c_2, c_3, c_4 all meet R_1 in $\{0, 1\}$, and $c_1 = 0$. Hence $|a \cap R_1|$ is even for every head $v(6, a)$ or $v(18, a)$ of $H_1, \dots, H_4$, and since ρ has an even number of minus signs, Lemma 4.2 gives $\langle h, 4\rho \rangle \leq 16$.

(c) $H_5, \dots, H_8$ against R_2: the same argument. The positions of R_2 outside O_{21} are 3 and 7. The basis words of K_2 meet $R_2 = \{0, 3, 7, 19\}$ in $\{0, 3, 7, 19\}$, $\{3, 7\}$, $\{0, 3, 7, 19\}$, $\{0, 3\}$ and $\{0, 3\}$, so every word of K_2 meets R_2 evenly; the representatives of γ'_3 and γ'_7 meet R_2 in $\{0, 3, 19\}$ and $\{0\}$, an odd number of points, and the labels c'_5, c'_6, c'_7 meet it in $\{0, 19\}$, $\{0, 3\}$ and $\{0, 7\}$, while $c'_8 = 0$. So $|a \cap R_2|$ is odd for every head $v(3, a)$ or $v(7, a)$ of $H_5, \dots, H_8$, and ρ has an odd number of minus signs. □

The configuration. Fix two orthogonal vectors $p_1, p_2 \in \Pi_0$. The configuration consists of:

- the body and the lifts of the eight head sets on the eight triangles, assigned as described below: $192336 + 12672$ points;
- poles at the four vectors of $\Pi_0 \setminus \{\pm p_1, \pm p_2\}$;
- the *Q-layer*: 64 layer points $Y(4\sigma, w)$ with $w \in \Pi_+$ and $\sigma \in \{\pm 1\}^Q$ with an even number of minus signs, and 64 with $w \in \Pi_-$ and σ with an odd number;
- the *R-layer*: 16 layer points $Y(4\rho, \pm p_1)$ with $\rho \in \{\pm 1\}^{R_1}$ even, and 16 layer points $Y(4\rho, \pm p_2)$ with $\rho \in \{\pm 1\}^{R_2}$ odd.

So each of the 16 directions of $\Pi_+ \cup \Pi_-$ and each of $\pm p_1, \pm p_2$ carries a cross-polytope of eight layer points, and the remaining four directions of Π_0 carry poles: $160 + 4 = 164$ points in place of the 24 poles of [8].

Assignment. The Q-layer is compatible with every head set by Lemma 4.1, and the R-layer is what the assignment has to handle. We put H_7, H_8 on the two triangles perpendicular to p_1 and H_3, H_4 on the two triangles perpendicular to p_2 (Lemma 4.5). This is possible because no triangle is perpendicular to both: the tails perpendicular to p_1 and p_2 are the four vectors $(\pm e_k \pm e_l)/\sqrt{2}$ in the remaining two coordinates, which contain no triangle. The other four head sets go on the other four triangles in any order. By Lemma 4.6 the assignment then has the following property, which we call the *assignment property*: if a triangle is not perpendicular to p_i, then its head set H_j has $m_i(j) = 16$.

Theorem 4.7. *This is a kissing configuration of* $192336 + 12672 + 4 + 128 + 32 = 205172$ *points in* $\mathbb{R}^{28}$.

Proof. Pairs among the body and the lifts are covered by Proposition 2.1. The remaining pairs involve a pole or a layer point.

Poles. A tail and a vector of Π_0 have inner product at most $\frac{1}{\sqrt{2}}$, so a lift and a pole have inner product at most $\beta/\sqrt{2} = \frac{1}{\sqrt{6}}$. Two poles have inner product 0 or -1, and poles are orthogonal to the body.

Layer, body. Lemma 4.3(1).

Layer, lift. A tail t has $\langle t, w \rangle \leq \frac{1}{\sqrt{2}}$ for every $w \in \Pi_0 \cup \Pi_+ \cup \Pi_-$. For the Q-layer, $\langle h, 4\sigma \rangle \leq 16$ for every head, and Lemma 4.3(2) applies. For an R-layer point with direction

$\pm p_i$ and a lift on a triangle not perpendicular to p_i, the assignment property gives $\langle h, 4\rho\rangle \le 16$, and Lemma 4.3(2) applies again. On a triangle perpendicular to p_i we have $\langle t, w\rangle = 0$.

Layer, pole. By Lemma 4.3(3) the value is $\langle w, p\rangle/\sqrt{2}$. An Q-layer direction has inner product $\pm\frac{1}{2}$ with every vector of Π_0, which gives at most $\frac{1}{2\sqrt{2}}$. The R-layer directions $\pm p_1, \pm p_2$ are orthogonal to the four poles.

Layer, layer. We check the cases of Lemma 4.4. Two points of the same part (Q-layer in Π_+, Q-layer in Π_-, R-layer at $\pm p_1$, or R-layer at $\pm p_2$) have the same tetrad and the same parity, and their directions are equal or have inner product 0 or -1: case (P1) or (P2). Between Π_+ and Π_-, the directions have inner product $\pm\frac{1}{2}$ and the parities on Q differ: case (P2) or (P3). Between the Q-layer and the R-layer, the directions have inner product $\pm\frac{1}{2}$ and $|Q \cap R_i| = 2$: again (P2) or (P3). Between the two parts of the R-layer, the directions are orthogonal: case (P2). □

Apart from the finite checks on the printed assignments (Definition 3.2 and the disjointness of the two groups), this proof is by hand. The Q-layer, 128 points in place of 16 poles, needs only Lemma 4.1 and Q-bipartite matchings; the R-layer, 32 points in place of 4 poles, is a further gain of 28 at the cost of the two extra tetrads, Lemma 4.6 and the assignment property.

5. Dimensions 29 to 31

In these dimensions we remove all poles. The configuration is a *frame*, the body and the lifts together with structured points in dimensions 29 and 30, plus a set E_n of *computed points*: 63, 97 and 167 integer vectors $z \in \mathbb{Z}^n$ with $|z_i| \le 30000$, used as the unit vectors $z/|z|$. We found them by numerical optimization against the frame, one point at a time, and rounded them to integer vectors. They have no closed form, and we prove nothing about them beyond this: every pair of points of the configuration that contains a computed point was checked exactly, as described in Section 6, and the largest cosine among these pairs is below 0.4999 in each dimension. The points are listed with the accompanying data. They sit where the poles were: the tail carries a large part of the squared norm of every computed point, at least a third in dimension 29 and at least half in dimensions 30 and 31, and in dimension 29 every tail points near a deep hole of the D_5 root lattice, a direction $s/\sqrt{5}$ with $s \in \{\pm 1\}^5$.

Dimension 29. The tails are $(\pm e_i \pm e_j)/\sqrt{2}$ in an orthonormal basis $e_1, \dots, e_5$ of $\mathbb{R}^5$, in 12 triangles and 2 antipodal pairs. Let $Q = \{0, 1, 10, 19\}$ as before. Four of the five octads through Q are

$$O_{11} = Q \cup \{3, 7, 14, 20\}, \qquad O_{21} = Q \cup \{6, 18, 21, 22\},$$
$$O_{41} = Q \cup \{4, 12, 16, 17\}, \qquad O_{51} = Q \cup \{5, 9, 11, 15\},$$

in the trios of K_1, K_2, K_4 and K_5. We use 4, 4, 4 and 2 head sets $H(\gamma, M)$ on them, with translates of one assignment on each octad and Q-bipartite matchings M; matchings on different octads through Q share no pair, because two such octads meet only in Q. The 14 head sets are pairwise disjoint (finite check), and every head h has $|\langle h, 4\sigma\rangle| \le 16$ for all $\sigma \in \{\pm 1\}^Q$ by Lemma 4.1.

The structured points are 144 layer points on Q, in 18 cross-polytopes, with directions in

$$\mathcal{V} = \{\pm e_1, \dots, \pm e_5\} \quad \text{and} \quad \mathcal{W} = \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_4 \pm e_5) \text{ with an even number of minus signs}\}.$$

The layer points are the 80 points $Y(4\sigma, w)$ with $w \in \mathcal{V}$ and σ even, and the 64 points with $w \in \mathcal{W}$ and σ odd. A tail has inner product in $\{0, \pm\frac{1}{\sqrt{2}}\}$ with each $w \in \mathcal{V}$, and in $\{0, \pm\frac{1}{2\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\}$ with each $w \in \mathcal{W}$. Two vectors of $\mathcal{V}$, or two of $\mathcal{W}$, have inner product 0 or

-1; a vector of $\mathcal{V}$ and one of $\mathcal{W}$ have inner product 0 or $\pm\frac{1}{2}$. The frame is the body, the lifts and these 144 points; the configuration is the frame and E_{29}.

Dimension 30. The tails are the 72 roots of E_6 in 24 triangles, with the 24 head sets of Table 4 on O_{12}, O_{13}, O_{21} and O_{23}. A unit vector $y \in \mathbb{R}^6$ is a *minuscule direction* if $\langle t, y\rangle \in \{0, \pm\frac{\sqrt{6}}{4}\}$ for every tail t; these are the directions of the 54 weights in the two minuscule orbits of E_6, which point at the deep holes of the E_6 root lattice. For $u \in \Lambda$ with $|u|^2 = 48$ and a unit vector $y \in \mathbb{R}^6$, the *hole point* with data (u, y) is

$$x(u, y) = \Big(\alpha\frac{u}{\sqrt{48}},\ \beta y\Big),$$

a point with the proportions of a lift. Like a layer point, a hole point is a lift of a lattice vector toward a deep hole of the tail lattice, but with $|u|^2 = 48$, so that it has the proportions of a lift. The frame is the body, the lifts and 72 hole points $x(u, y)$, for 72 pairs (u, y) with 36 distinct vectors u of shape $(\pm4, \pm2^8, 0^{15})$ and 18 distinct minuscule directions y, chosen by computer and listed with the accompanying data. They take the place of the 72 poles of [8]; the computed points E_{30} were found against this frame. Two facts about them were checked by computer:

(C1) for every hole point $x(u, y)$, every triangle not perpendicular to y carries a head set with $\langle u, h\rangle \le 16$ for all its heads h (a triangle is perpendicular to y if all three tails are);

(C2) the hole points have pairwise cosines at most $\frac{1}{2}$.

Dimension 31. The tails are the 126 roots of E_7 in 42 triangles, with the 42 head sets of Table 4. The frame is the body and the lifts, and the configuration is the frame and E_{31}.

Theorem 5.1. *These are kissing configurations of* $189168 + 21120 + 144 + 63 = 210495$, $183888 + 38016 + 72 + 97 = 222073$ *and* $174384 + 66528 + 167 = 241079$ *points in* $\mathbb{R}^{29}$, $\mathbb{R}^{30}$ *and* $\mathbb{R}^{31}$.

Proof. Pairs among the body and the lifts are covered by Proposition 2.1, and every pair with a computed point was checked exactly. It remains to check the pairs with a layer point in dimension 29 and with a hole point in dimension 30.

Dimension 29. Layer, body: Lemma 4.3(1). Layer, lift: every head has $\langle h, 4\sigma\rangle \le 16$ and every tail has $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$, so Lemma 4.3(2) applies. Layer, layer: all layer points lie on Q. Two points with directions in $\mathcal{V}$, or two with directions in $\mathcal{W}$, have the same parity and directions that are equal or at inner product 0 or -1: case (P1) or (P2) of Lemma 4.4. A point with direction in $\mathcal{V}$ and one with direction in $\mathcal{W}$ have opposite parities and directions at inner product 0 or $\pm\frac{1}{2}$: case (P2) or (P3).

Dimension 30. Hole, body: the inner product is $\alpha\langle u, b\rangle/\sqrt{48 \cdot 32}$, and $\langle u, b\rangle \le 24$ by (2.1), so it is at most $\sqrt{\frac{2}{3}} \cdot \frac{24}{16\sqrt{6}} = \frac{1}{2}$. Hole, lift: the inner product is $\frac{2}{3}\frac{\langle u,h\rangle}{16\sqrt{6}} + \frac{1}{3}\langle t, y\rangle$ with $\langle t, y\rangle \le \frac{\sqrt{6}}{4}$. On a triangle not perpendicular to y, (C1) gives $\langle u, h\rangle \le 16$ and the value is at most $\frac{2}{3\sqrt{6}} + \frac{\sqrt{6}}{12} = \frac{7\sqrt{6}}{36} < \frac{1}{2}$. On a perpendicular triangle $\langle t, y\rangle = 0$ and $\langle u, h\rangle \le 24$ by (2.1), so the value is at most $\frac{1}{\sqrt{6}}$. Hole, hole: (C2). □

Theorems 4.7 and 5.1 prove Theorem 1.1.

6. Computer verification

The proofs use the following finite facts, checked by computer in exact arithmetic: each coset assignment satisfies Definition 3.2, and the head sets in each dimension are pairwise disjoint; the facts (C1) and (C2) about the hole points in dimension 30; and in dimensions 29 to 31, that every pair containing a computed point has cosine at most $\frac{1}{2}$. The frames are determined by the text up to the choices we give with the accompanying data: the coset assignments and translation labels for dimensions 29 to 31, the partition of the tails into triangles, the assignment of head sets to triangles in dimensions 29 to 31, and the 72 hole points. In dimension 28 any partition into triangles works, and the assignments, labels and matchings are printed in Section 4.

As an independent check, the accompanying data list every point of each configuration as a vector with coordinates in $\mathbb{Z}[\sqrt{2},\sqrt{3}]$, and an accompanying program, written independently of the programs that built the configurations, checks all $N(N-1)/2$ pairs of each list: 21 047 672 206, 22 153 967 265, 24 658 097 628 and 29 059 421 581 pairs in dimensions 28 to 31. For each pair it decides (2.2), that is $\langle x,y\rangle \le \frac{1}{2}|x||y|$, by exact sign tests in $\mathbb{Z}[\sqrt{2},\sqrt{3}]$ with 128-bit and 256-bit integers, and it confirms that no two points have the same direction. It uses no structure of the configurations, no symmetry reduction and no floating point in any decision. It reports no violation in any of the four configurations, and in each the largest cosine is exactly $\frac{1}{2}$.

AI usage statement

GPT-6 Astra Pro, Claude Opus 5.5 and Claude Fable 5.1 were used to find these constructions, prepare supplementary data and draft this paper. The manuscript was proofread and lightly revised by the author. The author welcomes detailed questions regarding the use of AI in this work.

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Data availability

The coordinates of the four configurations, including the computed points, the coset assignments and the hole points, and an exact checking program accompany this paper. They can be found at `https://github.com/Felpix-Studios/kissing-number`.

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